THE NOVA RESEARCH CURRICULUM

Physics

Grade 8 · Term 1

Student edition

Outcome: PH-8.1 reflection, refraction, mirrors and colour.

Lesson notes

- **Reflection (Weeks 1–4):** law of reflection: angle of incidence = angle of reflection, measured from the normal. Plane mirror image: same size, upright, virtual, as far behind as the object is in front, left and right swapped. Concave mirrors (torches, shaving mirrors) and convex mirrors (vehicle side mirrors, shop security). Ray-diagram practical with a ray box or torch and slit.

  ▶ **Try it on NovaSchool:** Ray box: reflection and refraction — open **Simulations** on your dashboard.

- **Refraction (Weeks 5–8):** light bends when it changes speed between materials; into a denser material it bends towards the normal. A pencil looks bent in water; a pool looks shallower. Convex lenses focus light (magnifying glass, eye); concave lenses spread it.
- **Colour (Weeks 9–12):** a prism splits white light into a spectrum (red, orange, yellow, green, blue, indigo, violet). A red object reflects red and absorbs the rest; white reflects all; black absorbs all. Rainbows.

  ▶ **Try it on NovaSchool:** A prism splits white light — open **Simulations** on your dashboard.

Workbook pages

Page	Task
1–2	Ray-diagram practical: incidence and reflection table.
3	Draw refraction through a glass block.
4	Colour filters and objects table.

Question bank

#	Outcome	Question	Your answer
1	PH-8.1	The angle of incidence is 35°. The angle of reflection is ...	55° / 35° / 70° / 90°
2	PH-8.1	Angles of incidence and reflection are measured from the ...	mirror / normal / floor / light source
3	PH-8.1	Vehicle side mirrors are ...	plane / concave / convex / glass blocks
4	PH-8.1	Light going from air into water bends ... the normal.	away from / towards / along / it does not bend
5	PH-8.1	A magnifying glass uses a ... lens.	concave / convex / flat / coloured

#	Outcome	Question	Your answer
6	PH-8.1	A green leaf looks green because it … green light.	absorbs / reflects / makes / blocks
7	PH-8.1	A plane-mirror image is upright and the same size. True or false.	True / False
8	PH-8.1	Why does a swimming pool look shallower than it is?	_____